CONFIDENTIAL DISCUSSION PROPOSAL

National Digital Addressing System Proposal

A government-owned national addressing infrastructure program for Equatorial Guinea — giving every home, building, business, institution, and service point an official physical and digital address.

Prepared for the Ministry of Transportation at the request of the Vice President

Purpose: to establish a national address registry and geospatial service layer that supports postal delivery, emergency response, planning, logistics, utilities, digital public services, and national modernization.

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FOUNDER'S NOTE

A patriotic infrastructure proposal for national order, visibility, and service delivery

Every serious modern state needs a trusted way to identify places. Without a reliable address system, homes stay difficult to find, services stay hard to deliver, and national planning remains slower, more manual, and less accountable than it should be.

> **Roads, homes, public buildings, businesses, and future cities already exist. What is missing is a shared national location language that government, citizens, emergency teams, couriers, utilities, and investors can all trust and use.**

This proposal is not about street signs alone. It is about building a national operating layer for location identity: one authoritative system for house numbering, street naming, building registration, geolocation, postal coding, and searchable address verification.

A country that wants stronger logistics, better emergency response, cleaner records, more efficient public services, and a more investment-ready business environment should not leave addresses informal or fragmented. Addressing is foundational infrastructure.

Equatorial Guinea has the opportunity to do this with dignity and discipline: start with a controlled pilot, define standards correctly, keep government ownership of the registry, and scale in phases with auditability and operational realism.

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Respectfully submitted,

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EXECUTIVE SUMMARY

Not a map project — a national address operating layer

The recommended program is a phased National Digital Addressing System led by government and designed to become the authoritative source of location identity across Equatorial Guinea.

For citizens

- Know and share a clear official address.
- Receive deliveries, utility visits, and emergency response more reliably.
- Verify and correct address records through a controlled channel.

For ministries and municipalities

- Maintain one trusted registry instead of fragmented local lists.
- Track coverage, verification, gaps, and field progress.
- Connect addressing to permits, taxation, census, land, and service workflows.

For logistics and utilities

- Reduce failed deliveries and service visits.
- Plan routes, zones, and dispatch with verified coordinates.
- Integrate approved address APIs into operations over time.

For national leadership

- Create measurable visibility into territorial coverage.
- Support urban planning, infrastructure decisions, and public-service readiness.
- Strengthen digital sovereignty through a government-owned registry.

Core proposal: approve a 9-month pilot covering standards, field capture, registry build, and public-service integration in priority zones of Malabo, then scale into a 3-year phased national rollout across Bata, Oyala/Ciudad de la Paz, and all provinces.

WHY NOW

The country needs a trusted way to identify places

Addressing affects far more than mail. A weak address system slows almost every service that depends on finding people, properties, or facilities correctly.

Public services

NEED LOCATION CERTAINTY FOR PERMITS, INSPECTIONS, NOTICES, AND RESPONSE.

Emergency response

NEEDS FAST, UNAMBIGUOUS BUILDING AND ROUTE IDENTIFICATION.

Utilities

NEED PRECISE SERVICE POINTS FOR INSTALLATION, METER RECORDS, AND MAINTENANCE.

Commerce and logistics

NEED TRUSTED ADDRESSES FOR DISPATCH, DELIVERY, AND RECONCILIATION.

Planning and census

NEED CONSISTENT TERRITORIAL UNITS AND VERIFIED BUILDING RECORDS.

Future cities

SHOULD NOT INHERIT PAPER-FIRST ADDRESS CONFUSION.

What this means

- If addressing remains informal, every digital service built on top of it becomes weaker and more expensive to operate.
- A national address registry should be treated as foundational infrastructure, like roads, cadastral records, and public-service directories.
- A mobile-first field workflow is mandatory because verification will happen on the territory, not only in offices.
- Data governance and verification rules matter as much as software. A bad registry at scale becomes a national headache.

NATIONAL OPPORTUNITY

Oyala / Ciudad de la Paz should be born address-ready

A future-facing administrative capital should not depend on vague directions, improvised landmarks, and disconnected local records. It should open with a formal location identity model already designed into its operations.

> **If a new capital represents the future of the nation, then every plot, building, road, ministry site, and service point inside it should already be identifiable, searchable, and service-ready by design.**

Digital-by-default public administration

Permits, inspections, public notices, and inter-ministry coordination work better when locations are formally identified.

Emergency and security readiness

Response teams need precise building references, route logic, and zone awareness, especially in expanding administrative districts.

Utility and infrastructure delivery

Water, power, telecom, roads, sanitation, and maintenance operations benefit immediately from address-ready records.

Investor and resident confidence

A city that is easy to navigate, verify, and service projects seriousness and administrative maturity.

PLATFORM MODEL

What the national system should contain

Layer	Capability	National value
National address registry	Authoritative building, plot, street, and zone records with administrative hierarchy and verification state.	Provides a single national source of truth.
Geospatial map layer	Interactive map of roads, districts, buildings, parcels, landmarks, and service points.	Makes the registry usable in the field and in planning offices.
Field capture application	Mobile/tablet workflow for enumerators to capture buildings, photos, coordinates, naming proposals, and validation status.	Supports structured territorial rollout and offline collection.
Address code engine	Generates official address formats, short codes, printable labels, and signage references.	Keeps standards consistent across ministries and municipalities.
Operations dashboard	Coverage metrics, approval queues, duplicate detection, team productivity, verification backlog, and reporting.	Gives leadership visibility and operational control.
Citizen and agency access	Search, verify, request correction, print proof of address, and approved API access for service operators.	Drives real-world adoption beyond the registry itself.
Integration gateway	Controlled interfaces for postal, utilities, emergency response, tax, land, planning, and logistics systems.	Prevents the address system from becoming an isolated database.

NATIONAL ADVANTAGES

Why the whole country benefits

A good address system is quiet infrastructure: when it works, everything above it becomes easier to find, route, verify, inspect, deliver, and improve.

Better public-service delivery

Agencies can issue, inspect, route, and follow up with clearer location identity.

Faster emergency response

Ambulance, police, and fire services gain a more reliable operational reference model.

Stronger economic activity

Businesses, delivery operators, and suppliers spend less time failing to locate people and places.

Cleaner national records

Property, land, permits, service accounts, and census work gain a shared address backbone.

Tourism and navigation

Visitors understand how to reach hotels, offices, venues, and services with less friction.

Digital sovereignty

Government retains ownership and control of a strategic national dataset.

DELIVERY ROADMAP

Recommended phased rollout

Phase 1	**Standards + pilot design** Define address standards, governance model, data schema, signage rules, pilot zones, field methodology, and ministry operating model.	**3 months**
Phase 2	**Pilot field capture + registry build** Capture streets and buildings in pilot zones, verify naming/numbering, launch registry, and connect first government workflows.	**6 months**
Phase 3	**Malabo + Bata scale-up** Expand to priority urban districts, strengthen dashboards, and onboard utilities, emergency services, and postal workflows.	**9-12 months**
Phase 4	**National expansion** Roll out province by province with contractor control, QA, training, and approved integrations.	**12-24 months**
Phase 5	**Long-term operations** Maintain the registry, govern updates, support new developments, and evolve agency integrations.	**Ongoing**

GOVERNANCE AND SECURITY

This must be run as national trust infrastructure

Government ownership of the registry

- The core address database should remain under formal government control.
- External vendors may support delivery, but not own the national source of truth.

Cross-ministry steering model

- Transport, interior/local administration, planning, lands, postal, telecom, emergency response, and municipalities should align standards early.
- One lead sponsor is necessary, but the system serves many institutions.

Verification before publication

- No building record should become authoritative without defined validation rules.
- Duplicates, naming conflicts, and unverified coordinates must be controlled operationally.

Security and auditability

- Role-based access, change logs, backups, incident handling, and approval history are mandatory.
- Strategic location data should not drift into unmanaged personal spreadsheets or contractor silos.

Pilot model: What the first pilot should prove

- Government can define and approve one usable national address standard.
- Field teams can capture roads, buildings, and coordinates in a repeatable workflow.
- The registry can search, verify, and print official address records reliably.
- At least one ministry workflow, one emergency-response use case, and one utility/logistics use case can operate against the pilot data.
- Leadership can see measurable coverage, verification rate, backlog, and rollout cost signals before national expansion.

> Recommended pilot geography: priority zones of Malabo, one government-service-heavy district, one fast-growing urban area, and one controlled zone in Oyala/Ciudad de la Paz to prove future-city readiness.

LEGAL AND REGULATORY FRAMEWORK

The registry needs legal force, not only software

National authority model

The State should designate the lead authority that approves address standards, naming rules, numbering policy, and inter-agency usage obligations.

Municipal execution boundaries

Municipalities should participate in street naming, local validation, and maintenance reporting under a common national standard rather than inventing incompatible local systems.

Proof-of-address status

The program should define when an address record becomes valid for residence proof, utility onboarding, inspection, delivery, and administrative correspondence.

Appeals and corrections

A formal process should exist for disputes over names, numbering, duplicates, parcel conflicts, and resident correction requests.

- A decree, regulation, or equivalent mandate should define ownership of the standard, approval authority, update rights, and institutional obligations to use the registry.
- Street naming, building numbering, and district zoning rules should be documented with version control and publication discipline.
- The legal layer should clarify how the address registry interacts with cadastral, land, tax, census, and municipal records.
- No national rollout should happen without a clear rulebook for authority, appeal, and update responsibility.

PHYSICAL ROLLOUT AND SIGNAGE

The physical layer must be operationally real

Street and district signage

The program should define sign formats, reflectivity, materials, installation standards, maintenance cycles, and replacement workflow for damaged or missing signage.

Building numbering operations

Number assignment rules should handle compounds, corner plots, apartment blocks, commercial floors, informal clusters, and future subdivisions without collapsing into inconsistency.

Pilot-to-scale installation model

Digital registry launch can precede full national signage completion, but physical rollout sequencing should still be planned by zone, contractor, and inspection batch.

Field QA and acceptance

Every installed sign and numbered building cluster should be auditable against approved records, photo evidence, and field acceptance reports.

- Fabrication specifications should be standardized nationally to avoid fragmented designs and inconsistent quality.
- Installation contractors should work against survey-ready zone packets rather than ad hoc instructions.
- A replacement and vandalism-response workflow should be planned from the beginning, not after loss starts accumulating.
- Physical signage inventory should stay linked to registry IDs for traceability and future maintenance.

DETAILED INTEGRATION MODEL

The address system must plug into real operations

Postal transformation

Postal workflows should move from vague routing toward address-backed sorting, route planning, delivery confirmation, and service-coverage improvement.

Emergency dispatch

Police, fire, and ambulance teams should receive searchable address references, zone logic, and dispatch-grade location identifiers in their operational tools.

Utilities and field maintenance

Electricity, water, sanitation, telecom, and meter teams should link service points to verified address records to reduce failed visits and orphan records.

Land, tax, and planning systems

Cadastral, permitting, taxation, and planning workflows should reconcile parcels and buildings against the address backbone rather than maintaining separate place identities.

Citizen channels

Call centers, service counters, web lookup, printed proofs, SMS/WhatsApp notices, and assisted verification channels should all reference the same official record.

Private-sector enablement

Approved APIs and data-access contracts should support logistics, banks, insurers, e-commerce operators, and large employers without compromising registry control.

NATIONAL OPERATING MODEL

Software alone will not run the system

Field workforce model

Enumerators, supervisors, QA leads, municipal validators, and central registry operators should each have defined responsibilities, escalation paths, and daily reporting structure.

Training and certification

Field and office users should be trained on capture standards, verification rules, exception handling, and safe use of devices and data.

Zone-based rollout command

The country should be divided into manageable operational zones with clear readiness gates, progress reporting, and acceptance sign-off before scale expansion.

Service desk and registry operations

A national service desk should manage correction requests, duplicate investigations, access provisioning, support incidents, and controlled record updates.

- Device inventories, SIM/data plans, transport planning, and supervision ratios should be budgeted as operating realities, not hidden assumptions.
- A daily and weekly reporting cadence should measure capture volume, validation quality, exceptions, and blocked zones.
- Field teams should work from approved basemaps, task packs, and naming/numbering rules rather than informal personal judgment.
- The operating model should support both pilot execution and long-term national maintenance after rollout.

DATA QUALITY AND LIFECYCLE CONTROL

Bad records at scale become a national liability

Confidence and verification states

Every record should carry status markers such as captured, pending review, approved, disputed, corrected, retired, or superseded.

Duplicate and conflict handling

The platform should flag overlapping coordinates, conflicting names, suspicious numbering gaps, and disputed parcel/building relationships before publication.

Update governance

New developments, demolitions, rezoning, and administrative renaming events should feed a controlled update pipeline with audit history.

Revalidation cycle

The system should include periodic re-checks for fast-changing urban areas so address records stay useful instead of becoming stale after first launch.

- Quality assurance should use scorecards, photo review, sampling, and supervisor spot checks.
- Each province or city batch should clear acceptance thresholds before being declared operational.
- A national correction log should prevent quiet data drift and unmanaged local edits.
- Version history should support legal defensibility and operational trust.

INFRASTRUCTURE AND CONTINUITY POSTURE

A sovereign registry needs continuity planning

Hosting model choice

Government should choose deliberately between sovereign hosting, trusted local data-center operation, hybrid cloud, or another approved model based on control, resilience, and procurement reality.

Disaster recovery

Backups are not enough. The program should define recovery objectives, failover procedures, secure restore drills, and continuity responsibilities.

Access control and security hardening

Authentication, role boundaries, encryption, key custody, endpoint management, and logging policy should be treated as essential infrastructure controls.

Monitoring and audit readiness

System health, sync failures, suspicious edits, API usage, field-upload backlog, and outage events should be observable in near real time.

- The registry should not depend on unmanaged contractor laptops, private spreadsheets, or undocumented exports.
- Business continuity planning should cover both digital outages and field-operations disruption.
- Security review should include administrative abuse, API misuse, device loss, and unauthorized record manipulation.
- Operations should be measurable enough for ministerial oversight and future audits.

PROGRAM RISKS AND MITIGATION

The hard parts should be named early

Institutional fragmentation

Risk: ministries or municipalities create parallel address lists. Mitigation: one national standard, formal mandate, and integration obligations.

Poor field consistency

Risk: teams capture inconsistent names, numbering, or coordinates. Mitigation: strong training, QA sampling, approval gates, and zone supervision.

Political or community disputes

Risk: contested street names, district boundaries, or numbering decisions delay adoption. Mitigation: defined appeal process and staged validation.

Stale registry after launch

Risk: records decay if no maintenance workflow exists. Mitigation: permanent operating model, change pipeline, and annual update funding.

INVESTMENT MODEL

Indicative planning budget for discussion

Workstream	Coverage	Budget range
Program discovery + standards	Legal/administrative model, address standard, data schema, pilot governance, operational blueprint.	$500,000 - $800,000
Platform architecture + registry build	Core registry, map layer, dashboard, search, verification workflow, and hosting foundation.	$1,200,000 - $1,800,000
Field capture tools + operations	Enumerator app, offline capture, devices/process readiness, QA workflow, team management.	$800,000 - $1,200,000
Pilot field execution	Pilot-zone survey, verification, signage planning data, and controlled rollout support.	$1,800,000 - $2,800,000
Integrations + agency onboarding	Postal, emergency, utility, planning, and reporting workflow onboarding for the pilot stage.	$500,000 - $900,000
Security, training, and stabilization	Role security, backups, monitoring, training, documentation, and initial support.	$400,000 - $700,000
Indicative 9-month pilot envelope	**Discussion budget before formal field survey and procurement refinement.**	**$5,200,000 - $8,200,000**

These figures reflect a top-band sovereign-infrastructure pricing posture for a serious government pilot, not a final procurement quote. Final cost depends on pilot geography, field-team size, existing base maps, hardware approach, signage scope, hosting/security posture, procurement structure, political delivery model, and ministry integration depth.

COMMERCIAL STRUCTURE

Suggested engagement structure

Option A — Strategic design + pilot leadership

Lead standards, architecture, product design, vendor orchestration, and pilot operating model while government controls implementation approvals.

Option B — Build + pilot execution support

Deliver the platform, coordinate the pilot system rollout, train teams, and support first agency integrations under a defined mandate.

Option C — Government task-force advisory

Support a ministry-led or presidency-backed task force with technical direction, scope definition, procurement shaping, and architecture review.

Assumptions and boundaries

- Government remains the owner of the national address registry and policy standards.
- Final administrative authority for naming and numbering must be defined clearly before large-scale rollout.
- Existing municipal, cadastral, or utility records may be incomplete and should be treated as inputs to verify, not truth to copy blindly.
- National signage manufacture and installation can be phased separately from digital registry launch if needed.
- This proposal does not assume that all agencies integrate on day one; the system should launch in controlled layers.
- Final procurement, legal review, and territorial policy decisions remain with government.

RECOMMENDATION

Recommended next step

Approve a structured discovery and pilot-design mandate first. The immediate objective should not be a premature nationwide rollout. It should be to define standards properly, choose pilot territories intelligently, build the national registry foundation, prove operational value in the field, and create evidence for scaled expansion with government confidence.

> **The right first win is not “we launched a map.” The right first win is “government can now identify, verify, search, and use addresses in real operations with confidence.”**